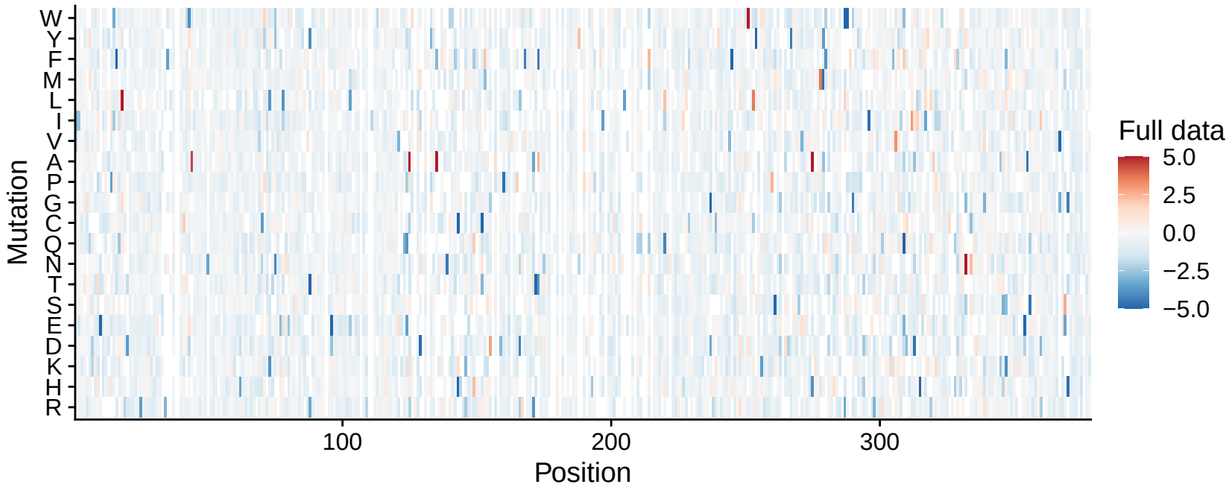
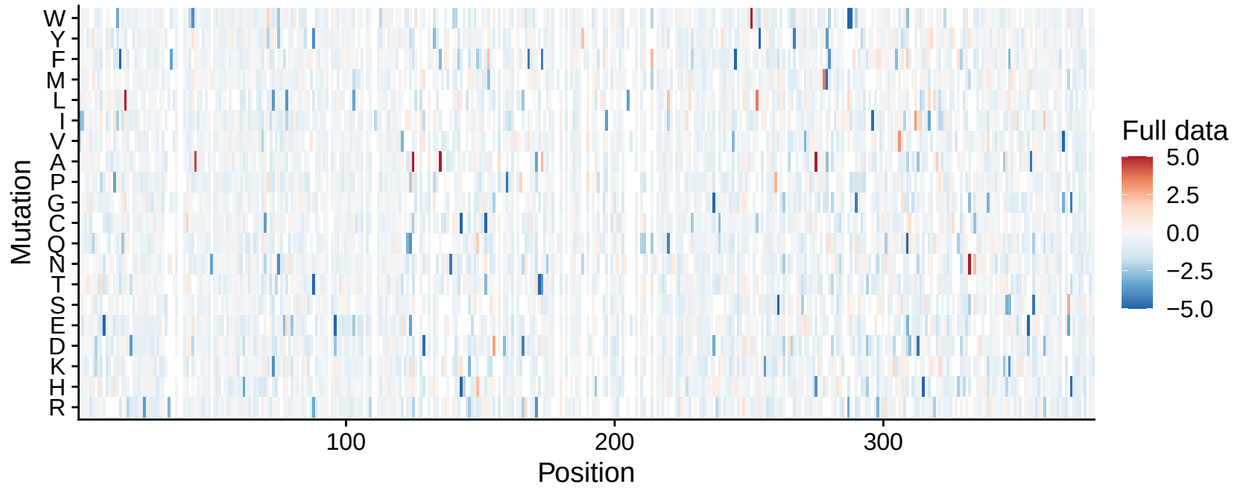
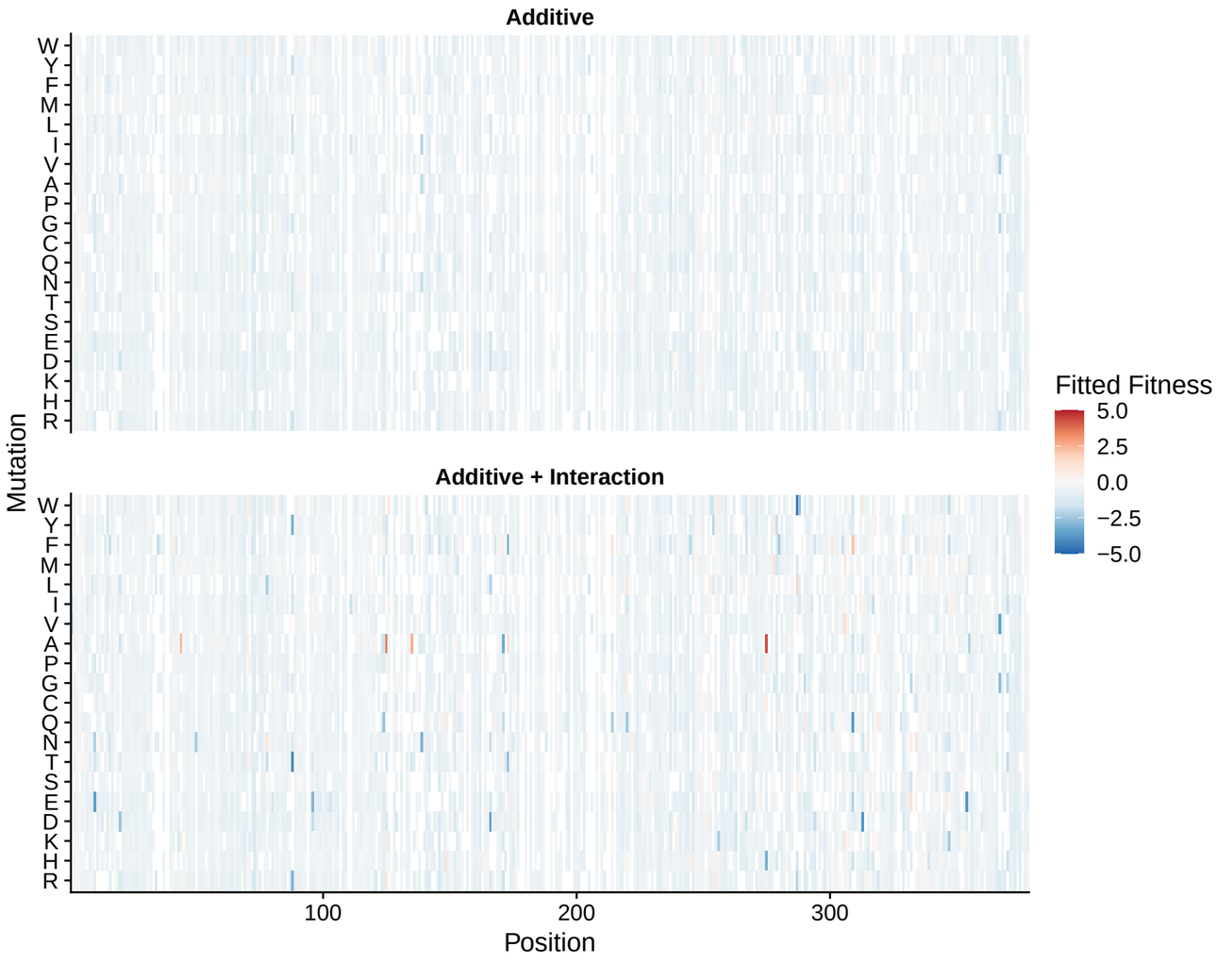
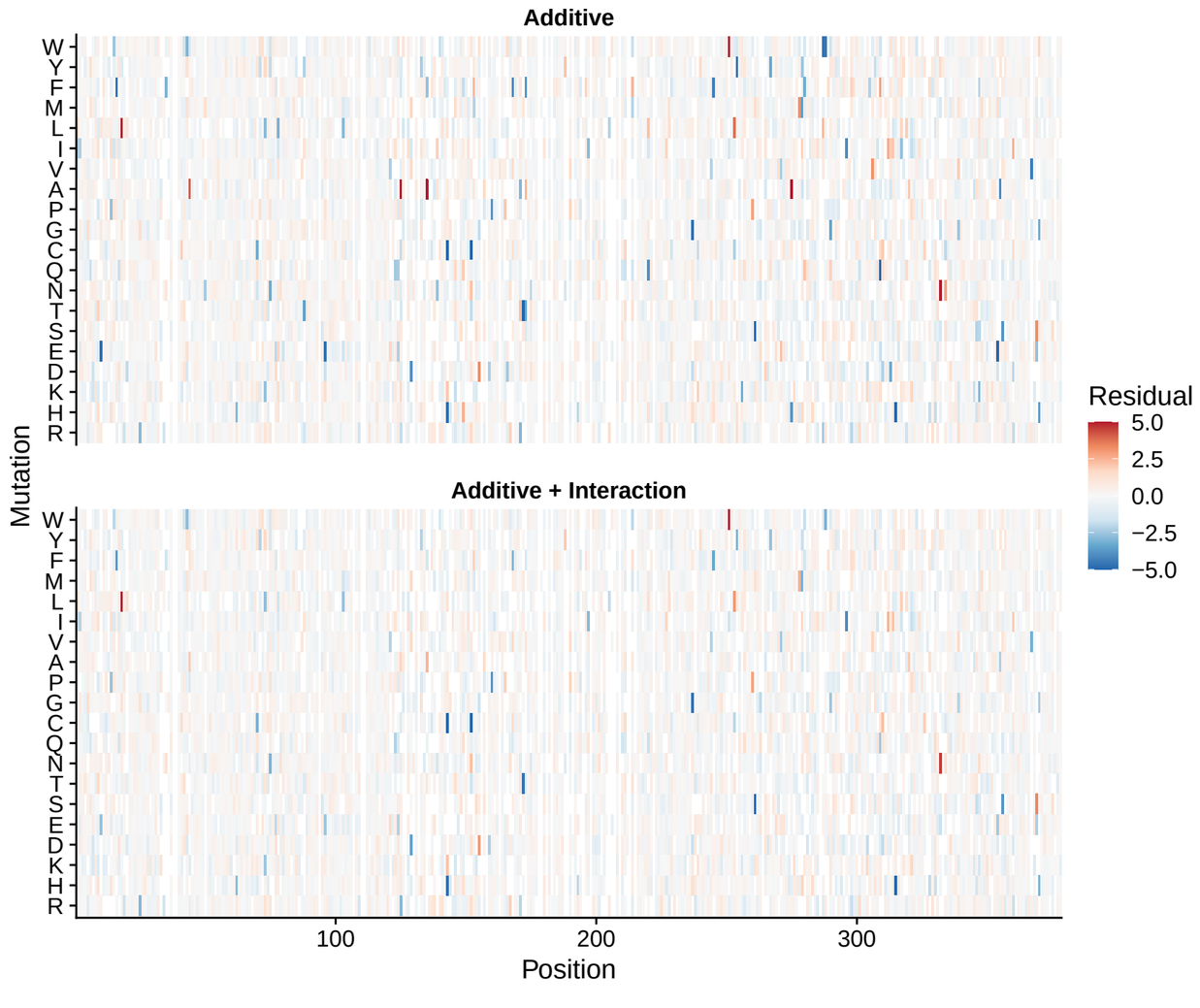

Full data
5.0
2.5
0.0
−2.5
−5.0
Mutation
W
Y
F
M
L
I
V
A
P
G
C
Q
N
T
S
E
D
K
H
R
100
200
300
Position


Full data
5.0
2.5
0.0
−2.5
−5.0
Mutation
W
Y
F
M
L
I
V
A
P
G
C
Q
N
T
S
E
D
K
H
R
100
200
300
Position


Additive
Additive + Interaction
Fitted Fitness
5.0
2.5
0.0
−2.5
−5.0
Mutation
W
Y
F
M
L
I
V
A
P
G
C
Q
N
T
S
E
D
K
H
R
100
200
300
Position


Additive
Additive + Interaction
Residual
5.0
2.5
0.0
−2.5
−5.0
Mutation
W
Y
F
M
L
I
V
A
P
G
C
Q
N
T
S
E
D
K
H
R
100
200
300
Position